

Python Coding for AI

Ch1: Python의 직관적 이해

Ch2: Python OOP

Ch3: What is Numpy & Why Numpy?

Ch4: What is Pandas & Why Pandas?

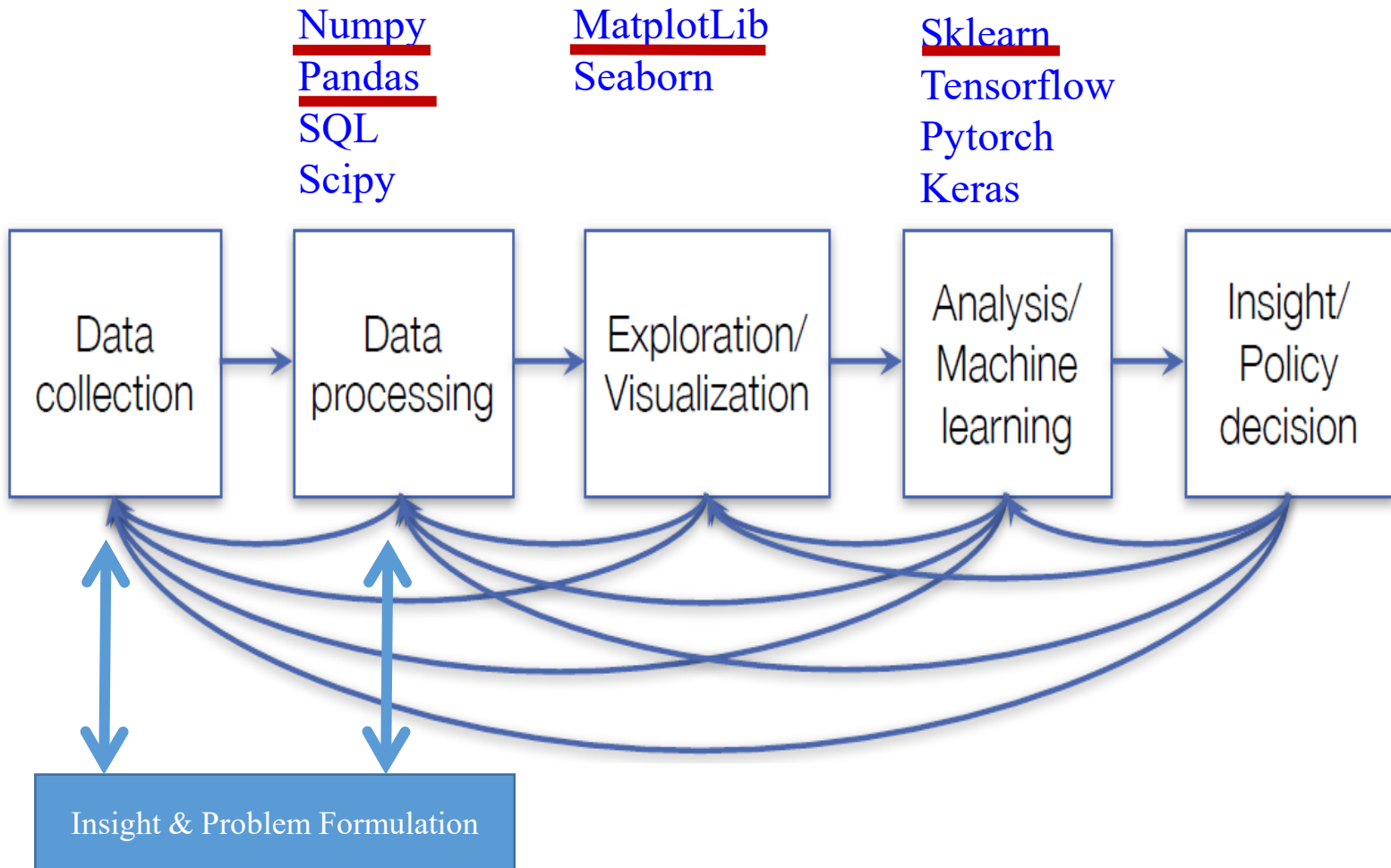
Ch5: What is Matplotlib & Why Matplotlib ?

Ch6: What is Sklearn & Why Sklearn ?

Ch7: What is Data Preprocessing & Why Data Preprocessing ?

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Major Steps of Data Analytics



“sklearn” module: Contents

- **Classification**: identify to which category an object belongs to
 - **Regression**: predict continuous-valued attribute
 - Linear regression, Logistic regression, Polynomial regression, etc ...
 - SVM, Decision Tree, Neural-Nets, K-Nearest-Neighbors, Bayesian, etc...
- **Clustering**: grouping of similar objects
 - K-means Clustering, Hierarchical Clustering, DB-Scan, etc ...
- **Model selection**: validate and choosing parameters and model
 - Cross Validation, Forward Selection, Backward Selection, etc ...
- **Data Preprocessing**: feature extraction & normalization
 - Feature extraction, One-hot Encoding, etc ...
- **Dimensionality reduction**: reducing number of variables
 - PCA, Feature Selection, etc ...
- **Datasets** : Iris Data, Diabetes Data, Boston Housing Data, etc



What is “sklearn” module?

- Scikit (SciPy Toolkit) –learn, SK-Learn
- Machine Learning Library for Python
- Designed to interoperate with Python numerical and scientific library **Scipy**
- Module Dependency
 - NumPy
 - Scipy
 - Matplotlib
- Open source (<http://scikit-learn.org>)
 - Initially developed by **David Cournapeau** as a “Google Summer of Code” project in **2007**
 - Still under active development
- Deep Learning은 다루지 않기로 결정!

scikit-learn	
	
Original author	David Cournapeau
Developer	Google Summer of Code project
Initial release	<u>June 2007</u> ; 18 years ago
Stable release	1.8.0 ^[1] / 10 December 2025; 2 months ago
Written in	Python, Cython, C and C++ ^[2]
Operating system	Linux, macOS, Windows
Type	Library for machine learning
License	New BSD License
Website	<u>scikit-learn.org</u>
Repository	github.com/scikit-learn/scikit-learn



scikit-learn

Machine Learning in Python

Getting Started

Release Highlights for 1.3

GitHub

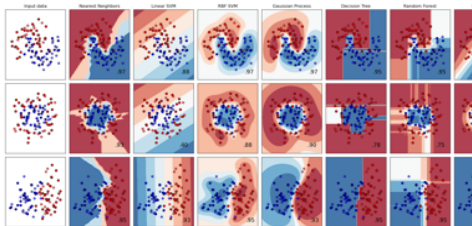
- Simple and efficient tools for predictive data analysis
- Accessible to everybody, and reusable in various contexts
- Built on NumPy, SciPy, and matplotlib
- Open source, commercially usable - BSD license

Classification

Identifying which category an object belongs to.

Applications: Spam detection, image recognition.

Algorithms: Gradient boosting, nearest neighbors, random forest, logistic regression, and more...



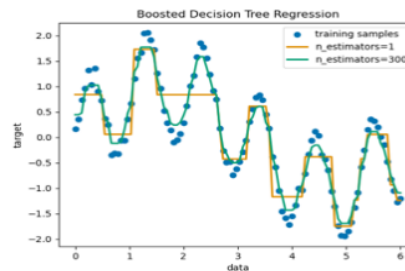
Examples

Regression

Predicting a continuous-valued attribute associated with an object.

Applications: Drug response, Stock prices.

Algorithms: Gradient boosting, nearest neighbors, random forest, ridge, and more...



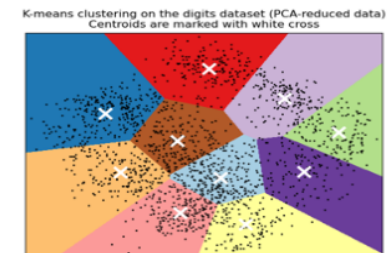
Examples

Clustering

Automatic grouping of similar objects into sets.

Applications: Customer segmentation, Grouping experiment outcomes

Algorithms: k-Means, HDBSCAN, hierarchical clustering, and more...



Examples

Dimensionality reduction

Reducing the number of random variables to consider.

Applications: Visualization, Increased efficiency

Algorithms: PCA, feature selection, non-negative matrix factorization, and more...

Model selection

Comparing, validating and choosing parameters and models.

Applications: Improved accuracy via parameter tuning

Algorithms: grid search, cross validation, metrics, and more...

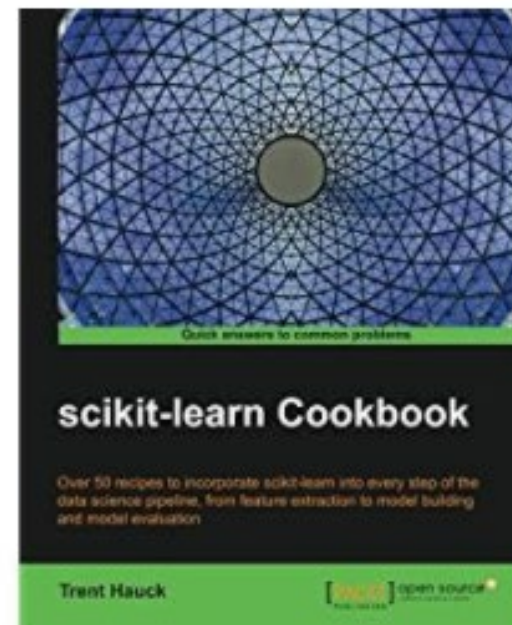
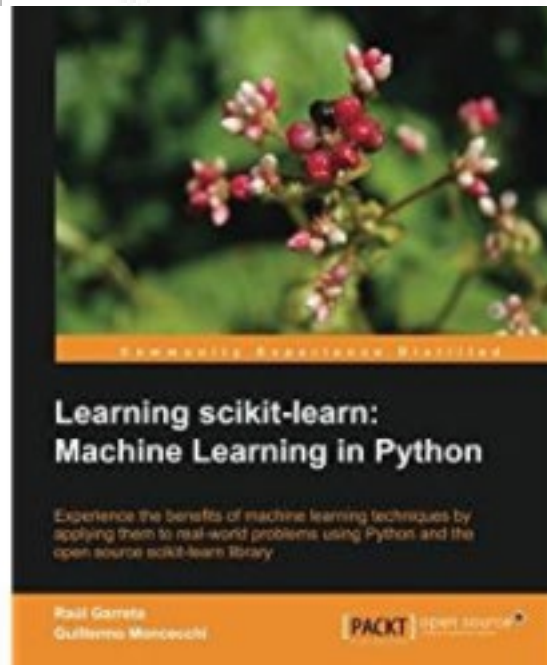
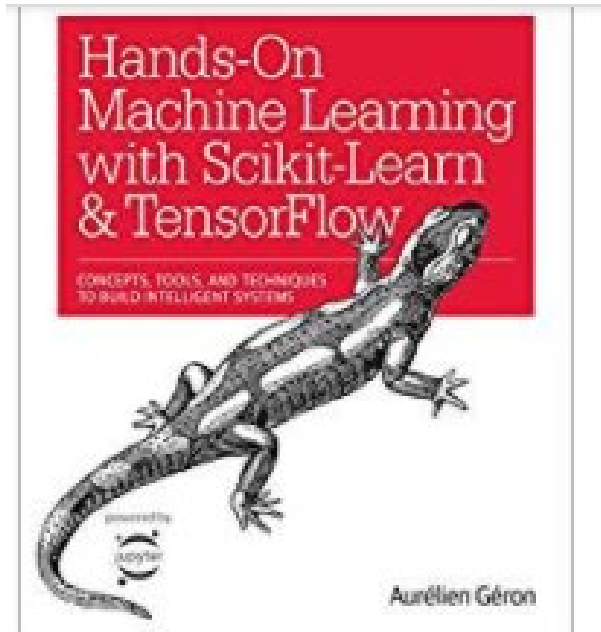
Preprocessing

Feature extraction and normalization.

Applications: Transforming input data such as text for use with machine learning algorithms.

Algorithms: preprocessing, feature extraction, and more...

Many SciKit-Learn Books



SK_Learn Module의 submodules

- `sklearn.datasets` submodule: 각종 data들의 load를 위한 함수들
- `sklearn.cluster` submodule: 각종 clustering algorithms들을 위한 functions과 classes
- `sklearn.linear_model` submodule: 각종 regression algorithms들을 위한 functions과 classes
- `sklearn.neighbors` submodule: 각종 nearest neighbors algorithms들을 위한 functions과 classes
- `sklearn.neural_network` submodule: 각종 neural network algorithms들을 위한 functions과 classes
- `sklearn.tree` submodule: 각종 decision tree algorithms들을 위한 functions과 classes
- 그외 30여개의 submodules

Sklearn.datasets module에 있는 다양한 load data 함수들

- `load_iris()`: 아이리스(iris) 꽃 데이터셋을 로드
- `load_digits()`: 손글씨 숫자 데이터셋을 로드
- `fetch_openml()`: OpenML 데이터 저장소에서 데이터셋을 로드
- `make_classification()`: 분류 문제를 위한 가상 데이터를 생성
- `make_regression()`: 회귀 문제를 위한 가상 데이터를 생성
- `load_boston()`: 보스턴 주택가격 데이터셋을 로드
- `load_diabetes()`: 당뇨병관련 데이터셋을 로드

`load_linnerud()` / `load_wine()` / `load_breast_cancer()` / `fetch_olivetti_faces()` /
`fetch_20newsgroups()` / `fetch_20newsgroups_vectorized()` / `fetch_lfw_people()` /
`fetch_lfw_pairs()` / `fetch_covtype()` / `fetch_rcv1()` / `fetch_kddcup99()` /
`make_multilabel_classification()` / `make_blobs()` / `make_gaussian_quantiles()` /
`make_hastie_10_2()` / `make_circles()` / `make_checkerboard()` / `make_moons()` /
`make_friedman1()` / `make_friedman2()` / `make_friedman3()` /
`make_low_rank_matrix()` / `make_sparse_coded_signal()` / `make_spd_matrix()` /
`make_swiss_roll()` / `make_s_curve()` / `make_sparse_uncorrelated()`

당뇨병 데이터의 regression 분석

Data				Label
Age	Sex	BMI	ABP	공복혈당
28	남	21.5%	80..120	141
72	여	19%	70..115	132
55	남	3%	60..90	93



Data Preprocessing

Data				Label
Age	Sex	BMI	ABP	공복혈당
0.038	0.051	0.062	0.022	141
-0.002	-0.045	-0.051	-0.026	132
0.085	0.051	0.044	-0.006	93

Regression Classifier Coding

```
import matplotlib.pyplot as plt
import numpy as np
from sklearn import datasets, linear_model
```

```
diabetes = datasets.load_diabetes()
diabetes_X = diabetes.data[ : , np.newaxis, 2]
```

데이터 전처리

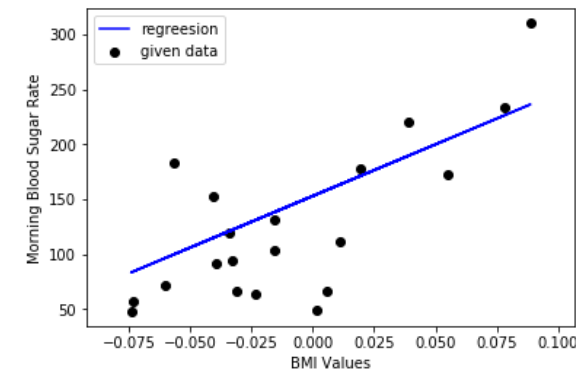
```
diabetes_X_train = diabetes_X[ :-20]
diabetes_X_test = diabetes_X[-20: ]
diabetes_y_train = diabetes.target[ :-20]
diabetes_y_test = diabetes.target[-20: ]
```

```
regr = linear_model.LinearRegression()
```

AI object 생성

```
regr.fit(diabetes_X_train, diabetes_y_train)
```

데이터 학습



```
plt.scatter( diabetes_X_test, diabetes_y_test, color = 'black', label = "test data")
plt.plot( diabetes_X_test, regr.predict(diabetes_X_test), \
         color = 'blue', label = "predicted points")
```

데이터 시각화

```
plt.xlabel("BMI Values")
plt.ylabel("Morning Blood Sugar Rate")
plt.legend()
```

AI Coding 을 위해 알아야할 Sklearn의 Details

- **Consistent API (일관된 인터페이스):** 모든 모델 학습에는 `fit()`, 예측는 `predict()`, 모델평가에는 `score()`라는 공통 메서드를 사용하여 알고리즘 교체가 용이
- **transform() vs fit():** 데이터 전처리는 `fit_transform()`, 모델 학습은 `fit()`을 사용
- **Pipeline (파이프라인):** `make_pipeline()`을 사용해 전처리와 모델 학습을 하나로 묶어 코드의 가독성을 높이고 교차 검증 시의 실수를 방지
- **Train/Test Split (데이터 분할):** `train_test_split( )`을 사용하면 분류 문제에서 클래스 비율을 유지하며 데이터를 나눔
- **Hyperparameter Tuning (Grid & Random Search):** `GridSearchCV`와 `RandomizedSearchCV`를 통해 최적의 파라미터 조합을 자동 탐색하고 성능을 극대화
- **Cross-Validation (교차 검증):** `cross_val_score( )`를 활용해 단일 분할의 우연성을 배제하고 모델의 일반화 성능을 객관적으로 평가
- **Model Evaluation Metrics (평가 지표):** `classification_report( )`, `confusion_matrix( )`, `roc_auc_score( )`등 문제 유형(분류/회귀)에 맞는 다양한 지표의 구사
- **Feature Selection (특성 선택):** 모델 성능에 기여도가 낮은 변수를 제거

김용대교수께서 3일간 기계학습을 하면서 전부 자세히
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